Everything in the universe is made up of essentially two things: matter and energy. Matter includes the physical things around us: the earth, the air you breathe, and your pencil. Energy is the ability to cause change or do work. You will learn what makes up matter and the various sources of energy. You will also learn how energy works within the food chain as well as the human effect on the food chain.

Matter and Energy

Read the information and trace the dotted words.

matter - all physical things around us

energy - ability to cause change or do work.

atoms - the smallest component of an element

kinetic energy - energy in use

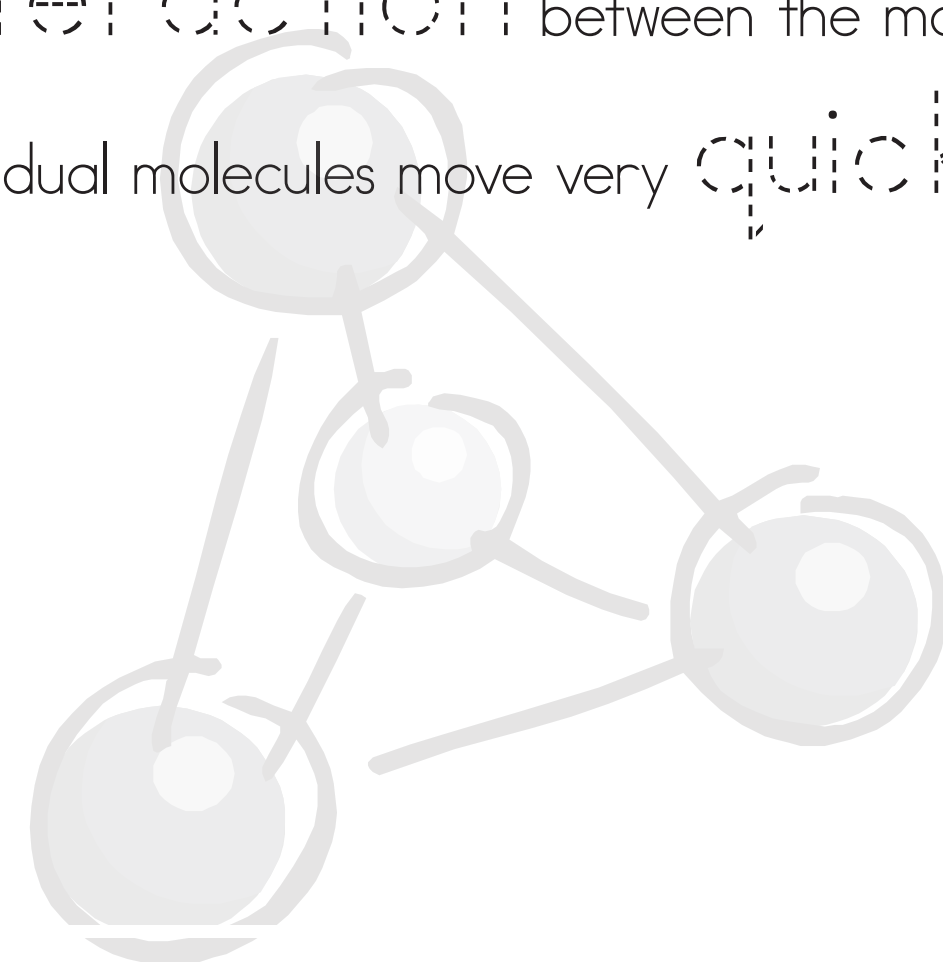
potential energy - energy that is stored

food chain - cycle by which energy is passed from one organism to another

universe - contains everything that exists in space

molecules - combination of atoms

Matter includes all the physical things around us. There are three fundamental states of matter: solid, liquid, and gas. In solids, molecules are packed tightly together. As energy is added, such as when heat is added to water, the amount of movement by each molecule is increased. The molecules slowly separate from each other and form the liquid state. As even more heat is added, the molecules separate further, until there is relatively little interaction between the molecules. At this point, the individual molecules move very quicky, forming gas.



M_____ includes all the physical things around us.

The three states of matter are S_____,

L_____, and G_____.

In solids, m_____ are tightly packed together.

As E_____ is added, the amount of

m_____ in the molecules

L_____, forming a L_____.

When more energy is added, the molecules S_____

further, forming a G_____.

Word Bank

gas	**matter**	**increases**	**liquid**
separate	**increases**	**liquid**	**gas**
energy	**liquid**	**molecules**	
movement	**matter**	**solid**	

All the physical things around us is called _____.

_____, _____, and _____

are the three fundamental states of matter.

When _____ is added, the amount of movement in

the _____ increase, forming a liquid state.

When even more energy is added, the molecules

_____ further, to the point where there is little

_____, forming gas.

Word Bank

matter	**separate**	**liquid**
solid	**interaction**	**molecules**
energy	**gas**	

Matter and Energy

Fill in the blank with the correct word.

All the physical things around us is called _____.
matter

_____, _____, and _____
solid **liquid** **gas**

are the three fundamental states of matter.

When _____ is added, the amount of movement in
energy

the _____ increase, forming a liquid state.
molecules

When even more energy is added, the molecules

_____ further, to the point where there is little
separate

_____, forming gas.
interaction

Word Bank

matter	**separate**	**liquid**
solid	**interaction**	**molecules**
energy	**gas**	

M	O	L	E	C	U	L	E	M
S	O	Y	Y	U	I	I	N	A
O	T	V	R	E	W	Q	E	T
L	F	W	E	D	S	U	R	T
I	G	A	H	M	J	I	G	E
D	T	T	N	M	E	D	Y	R
B	A	E	V	X	G	N	Z	A
H	E	R	G	A	F	D	T	S
P	H	Y	S	I	C	A	L	J

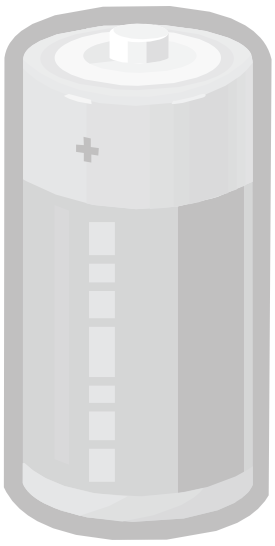
Word Bank

matter	**gas**	**heat**	**movement**
solid	**energy**	**water**	
liquid	**molecule**	**physical**	

Matter and Energy

Find the words in the word search.

Energy is the ability to cause change or do work. Forms of energy include light, heat, electrical energy, and mechanical energy, such as movement. There are two types of energy: potential and kinetic. Potential energy is energy that is stored. Kinetic energy is energy in use. For example, if you drop a ball from the top of a ladder, the ball possesses kinetic energy as it falls. Where did that come from? It was stored in the ball when the ball was suspended at the top of the ladder, and this stored energy is called potential energy.



Energy

Read the information and trace the dotted words.

E_____ is the ability to do work.

Some forms of energy are l_____, h_____,

electrical energy, and mechanical energy, such as

m_____. The two types of energy

are p_____ and k_____.

Potential energy is energy that is s_____.

Energy that is u_____ is called kinetic energy.

Word Bank

kinetic	**energy**	**movement**
potential	**heat**	**used**
stored	**light**	

Energy

Fill in the blanks and trace the dotted words.

1. What is the ability to change or do work?

energy matter water

2. What is energy that is stored?

kinetic potential light

3. What is energy that is used?

kinetic molecules matter

4. When a ball is dropped from the ladder, it has what kind of energy?

kinetic electrical potential

5. When a ball is suspended at the top of a ladder, it has what kind of energy?

kinetic heat potential

Energy

Circle the correct answer.

1. What is the ability to change or do work?

energy matter water

2. What is energy that is stored?

kinetic potential light

3. What is energy that is used?

kinetic molecules matter

4. When a ball is dropped from the ladder, it has what kind of energy?

kinetic electrical potential

5. When a ball is suspended at the top of a ladder, it has what kind of energy?

kinetic heat potential

Example:
H eather likes the summer
E xcept when it rains
A nd then she has to stay inside and watch
T elevision or do chores

E _____

N _____

E _____

R _____

G _____

Y _____

E _____

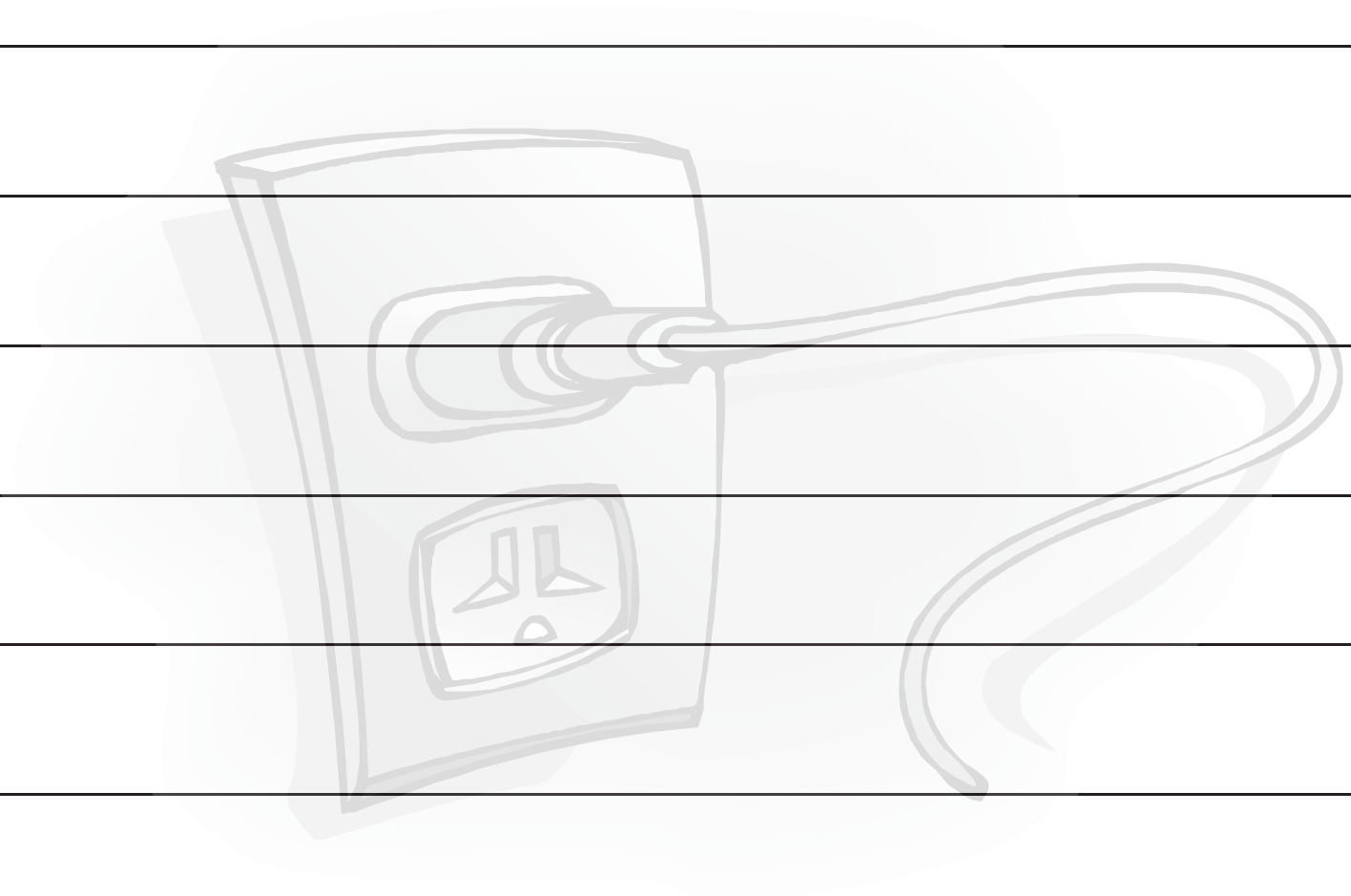
N _____

E _____

R _____

G _____

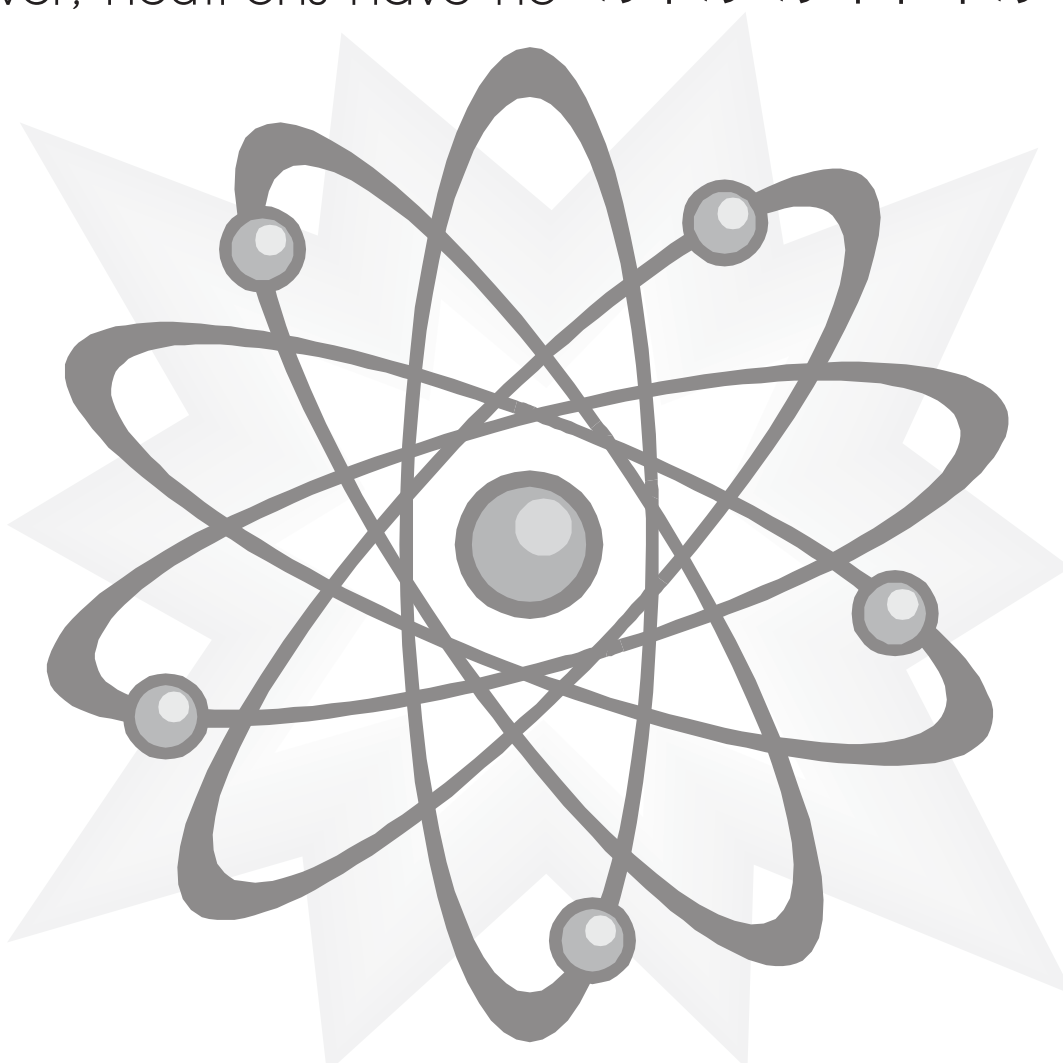
Y _____



Energy

Make an acrostic poem with the word "energy."

Atoms are small parts of elements that make up matter and cannot be broken down any further. Three different types of particles are electrons, protons, and neutrons. Electrons are tiny, very light particles that have a negative charge. Protons are much larger and heavier than electrons and have a positive charge. Neutrons are large and heavy like protons; however, neutrons have no electrical charge.



Atoms

Trace the dotted words.

A_____ are the particles of

e_____ that make up

m_____, and cannot be broken down any

further. There are three different particles in an atom:

e_____, p_____,

and n_____.

E_____ are tiny, very l_____

particles that have a n_____ charge.

P_____ are much larger and heavier than

electrons and have a p_____ charge.

N_____ are l_____

and heavier, but have no e_____

charge.

Word Bank

neutrons	**positive**	**atoms**	**electrons**
large	**light**	**elements**	
electrical	**large**	**matter**	
protons	**negative**	**substance**	

Atoms

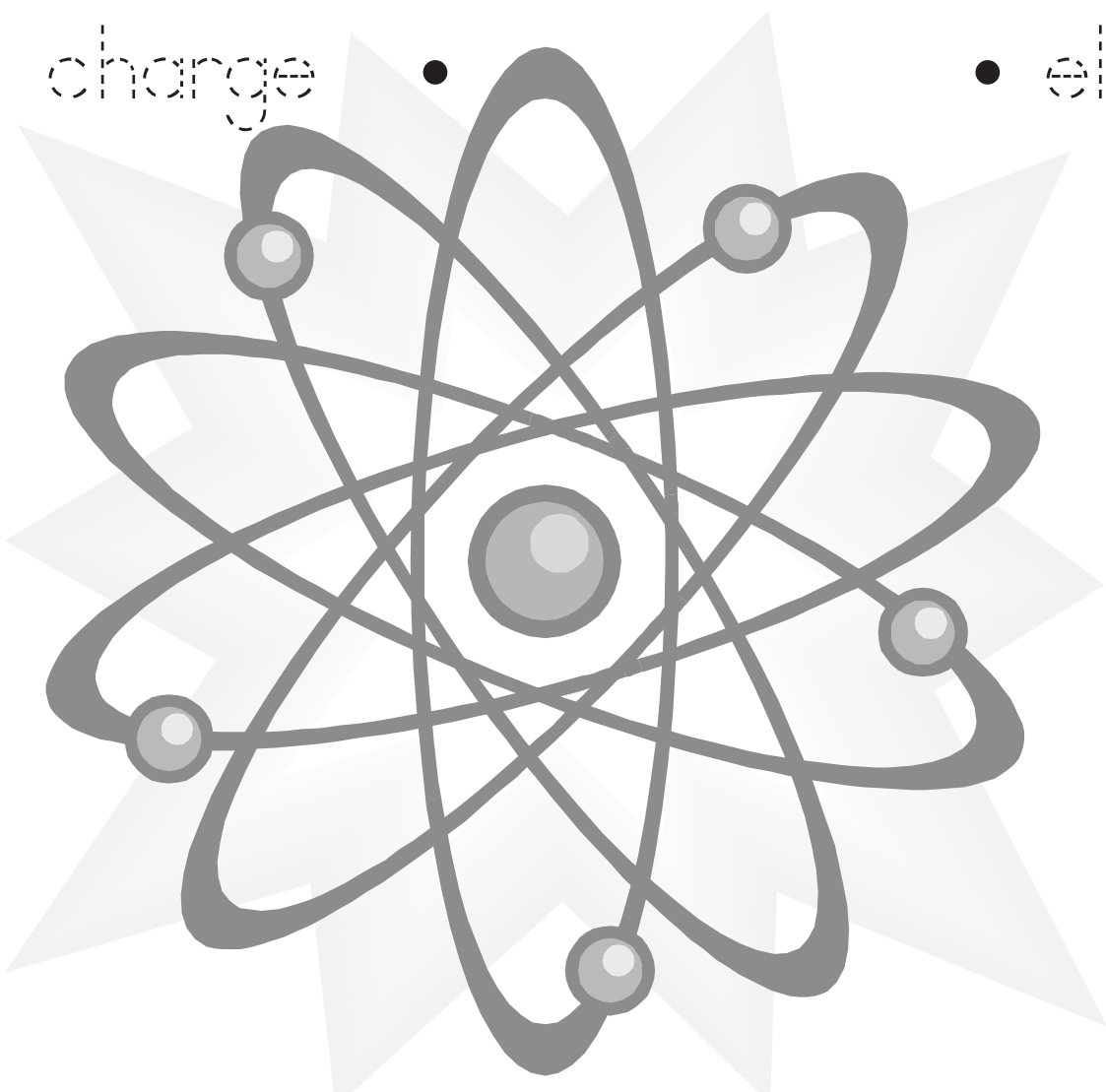
Fill in the blanks.

have a
negative charge • • atom

have a
positive charge • • protons

particles that
make up matter • • neutrons

have no
electrical charge • • electrons



Atoms

Draw a line to the correct word and trcae the dotted words.

have a
negative charge

have a
positive charge

particles that
make up matter

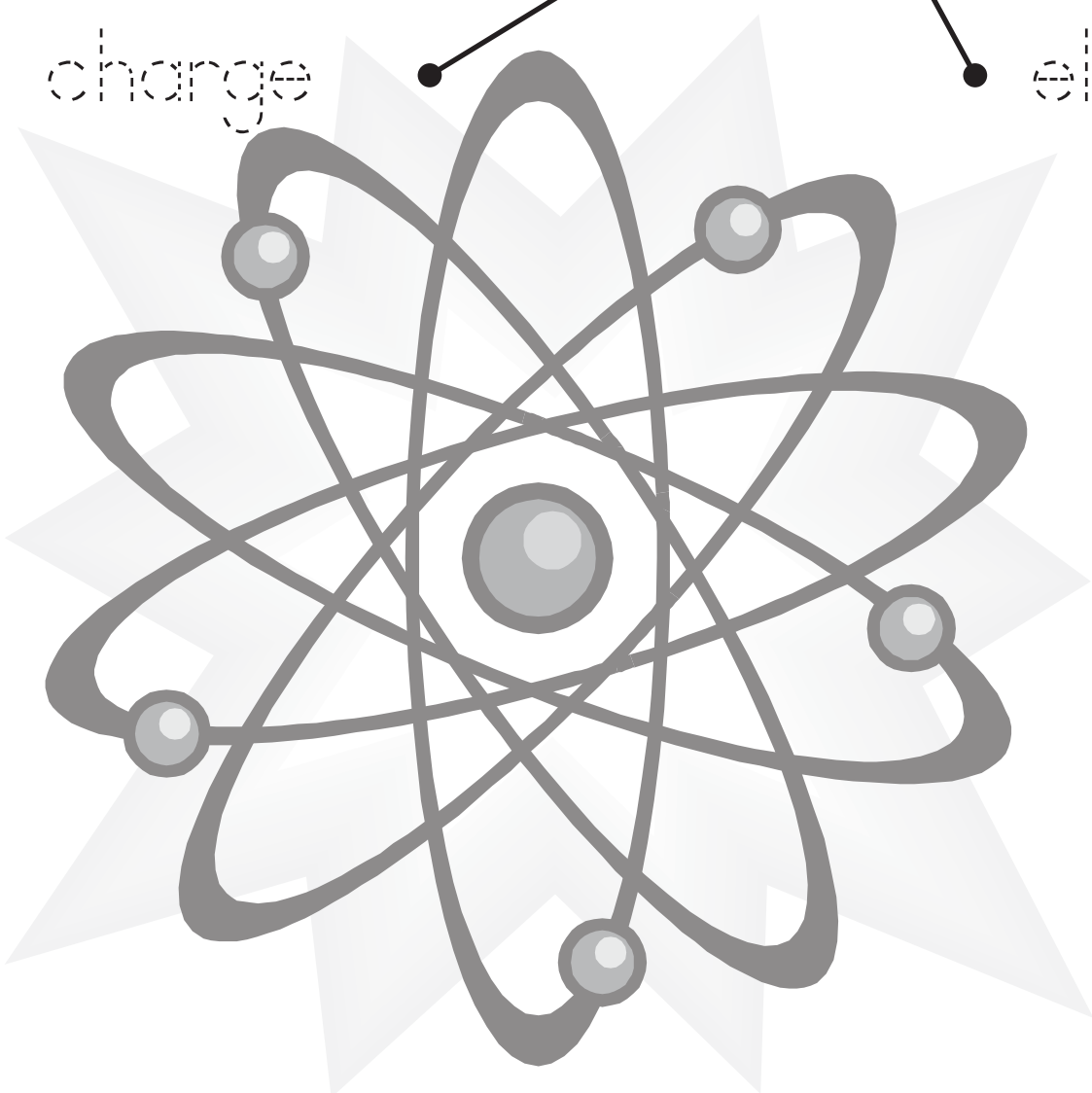
have no
electrical charge

atom

protons

neutrons

electrons



somta _____

onseelctr _____

oosnrpt _____

uenronts _____

eaiengtv _____

oiiepstv _____

prtclsaie _____

treamt _____

eeelmnts _____

Word Bank

elements	**neutrons**	**electrons**
matter	**negative**	**protons**
positive	**atoms**	**particles**

Atoms

Unscramble the words.

We use energy in everything we do, from making a jump shot to baking cookies. Sources of energy are divided into two groups: renewable and nonrenewable. Renewable energy is energy that can be replenished in a short period of time. Some examples of renewable sources include solar energy, which comes from the sun; wind energy, geothermal energy from inside the earth, biomass from plants, and ocean energy from the power of the waves. However, we get most of our energy from nonrenewable sources, which we are using up and cannot recreate in a short period of time. Oil, natural gas, and coal are just a few examples of nonrenewable sources. We use all these energy sources to generate the electricity we need to power our homes, businesses, schools, and factories.

Sources of Energy

Trace the dotted words.

Energy is divided into two groups: r_____

and n_____.

A r_____ source is a source that can be replenished in a short period of time.

A n_____ source is a source that we are using up and cannot recreate in a short period of time.

S_____ energy, wind, and geothermal energy are examples of renewable sources.

_____, natural gas, and C_____ are examples of nonrenewable sources.

Word Bank

renewable	**oil**
nonrenewable	**coal**
solar	

Sources of Energy

Fill in the blanks and trace the dotted words.

1. Sources of energy are divided into 2 groups: _____ and nonrenewable.

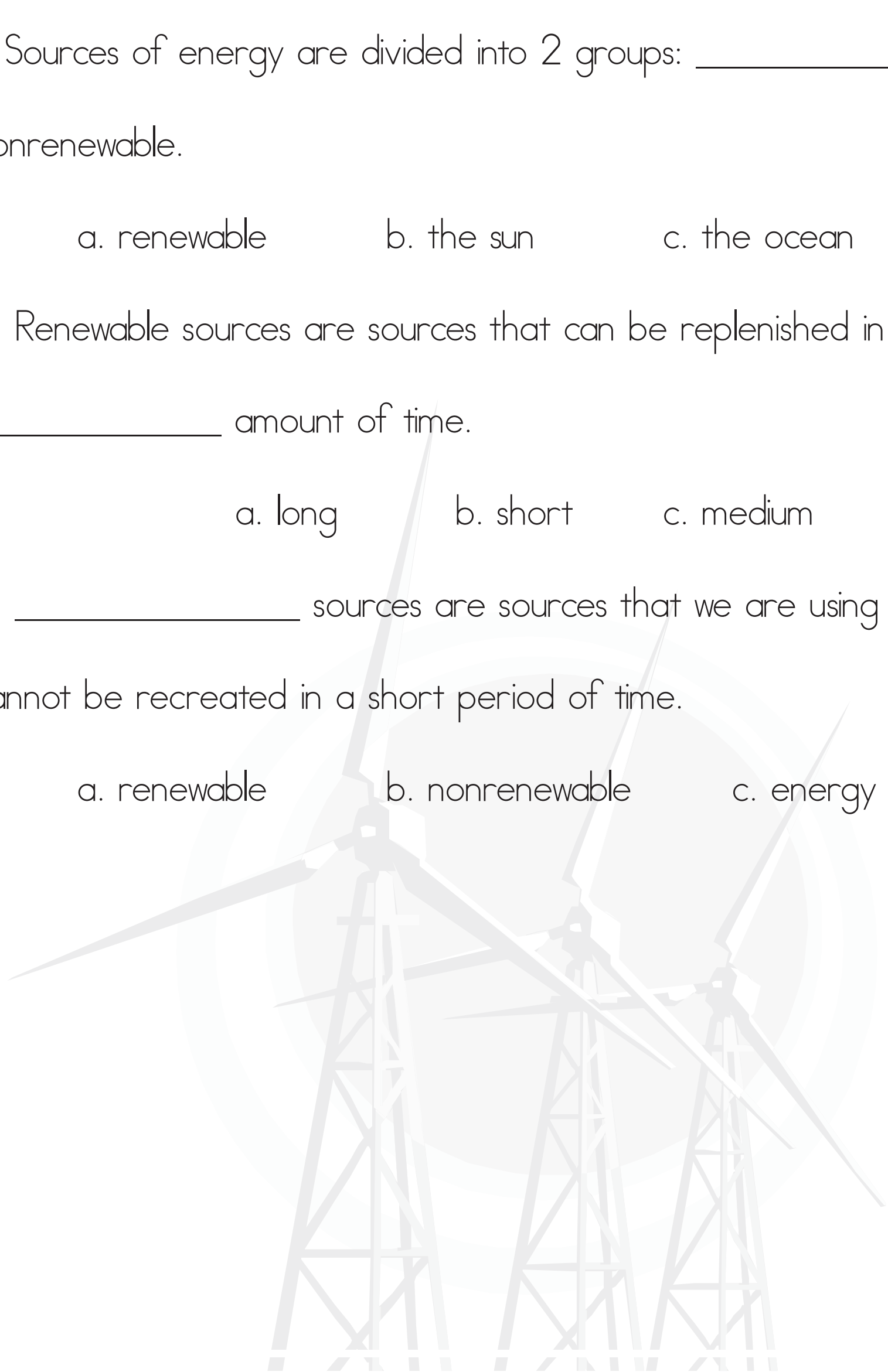
a. renewable b. the sun c. the ocean

2. Renewable sources are sources that can be replenished in a _____ amount of time.

a. long b. short c. medium

3. _____ sources are sources that we are using up and cannot be recreated in a short period of time.

a. renewable b. nonrenewable c. energy



Sources of Energy

Circle the correct answer.

1. Sources of energy are divided into 2 groups: _____ and nonrenewable.

a. renewable b. the sun c. the ocean

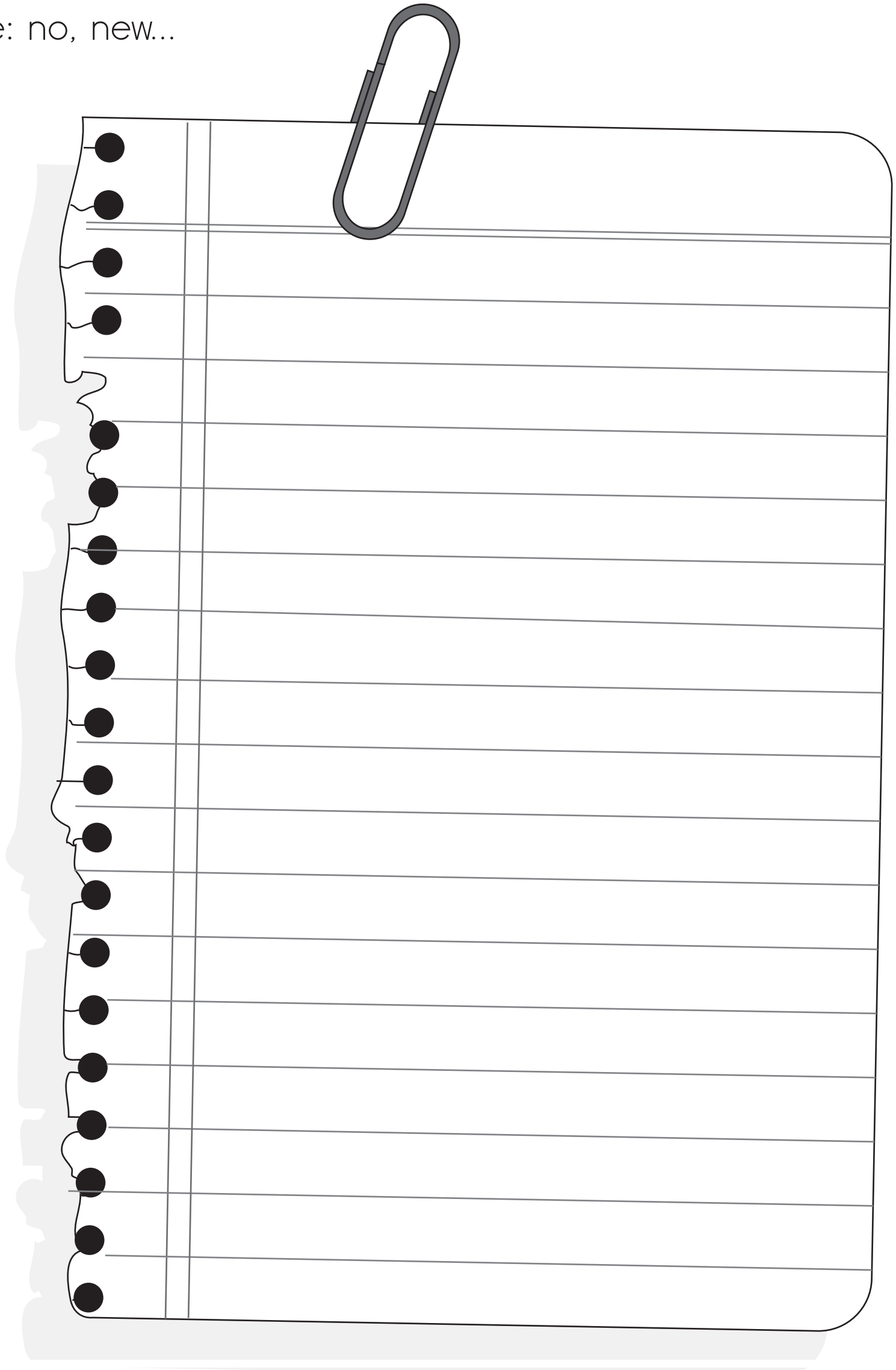
2. Renewable sources are sources that can be replenished in a _____ amount of time.

a. long b. short c. medium

3. _____ sources are sources that we are using up and cannot be recreated in a short period of time.

a. renewable b. nonrenewable c. energy

Example: no, new...



The food chain is a complex web where energy is passed from one organism to another. There are four main parts of the web. The first is the Sun, which provides energy for everything on the planet. Second are the producers, which are all the green plants, also known as autotrophs since they make their own food. Third are the consumers which include every organism that eats something else. Finally, decomposers, which are mainly bacteria and fungi that convert dead matter into gases such as carbon and nitrogen, to be released back into the air.

Mmmm! The Food Chain

Trace the dotted words.

The f_____ c_____ is a complex web where energy is passed from one organism to another.

The S_____ provides energy for everything on the planet.

All green plants, known as autotrophs, are called the p_____.

The c_____ include every organism that eats something else.

d_____ are mainly bacteria and fungi that convert dead matter into gases.

Word Bank

decomposers	**sun**
consumers	**chain**
food	**producers**

_____ a complex web where energy is passed from one organism to the next.

_____ provides energy for everything on the planet.

_____ all green plants.

_____ every organism that eats something else.

_____ bacteria and fungi that convert dead matter into gases.

a. sun

b. consumers

c. decomposers

d. food chain

e. producers

Mmmm! The Food Chain

Write the letter of the correct answer in the blank.

_____ a complex web where energy is passed
d
from one organism to the next.

_____ provides energy for everything on
a
the planet.

_____ all green plants.
e

_____ every organism that eats something
b
else.

_____ bacteria and fungi that convert dead
c
matter into gases.

a. sun

b. consumers

c. decomposers

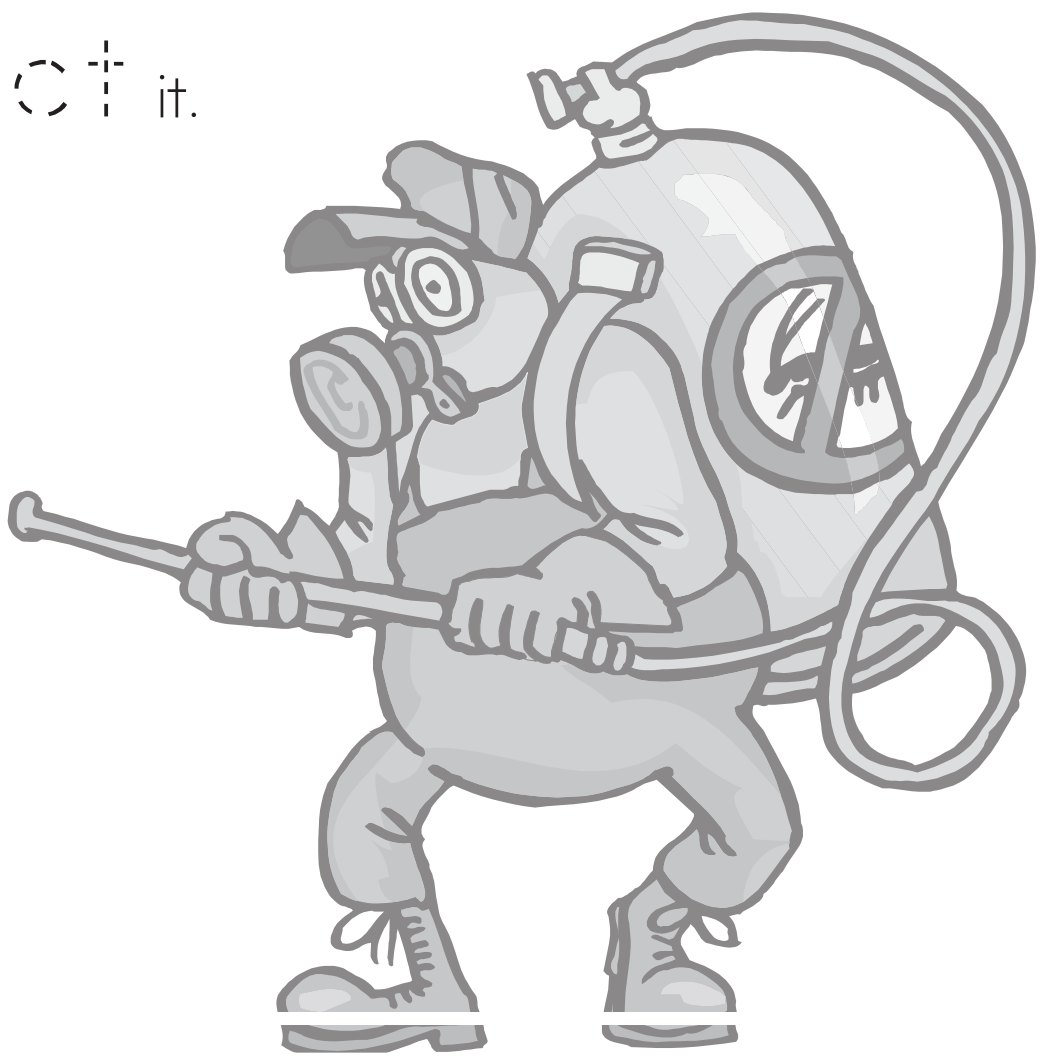
d. food chain

e. producers

Mmmm! The Food Chain

Draw an example of a consumer and the things it eats.

When we spray pesticides, chemicals used to kill insects, we put the food chain in danger. Breaking a link in the chain means all the organisms above that link are in danger of extinction. Hunting animals nearly to extinction can put everything above that animal in danger. Since the food chain provides energy that all living things must have in order to survive, it is imperative that we protect it.



Human Effects on the Food Chain

Trace the dotted words.

When we spray P_____, chemicals used

to kill I_____, we put the food chain in

d_____.

When a I_____ in the food chain is broken, we are

putting all O_____ above that link in

danger of D_____. Since the food chain

provides E_____ that all living things must have

to S_____, it is imperative that we

P_____ it.

Word Bank

protect	**organisms**	**danger**
energy	**pesticides**	**survive**
link	**insects**	**extinction**

Human Effects on the Food Chain

trace the dotted words.

1. _____ chemicals used to kill insects.

2. _____ when a link is broken, the animals above it are in danger of what?

3. _____ when we _____ animals nearly to extinction, we are putting the food chain in danger.

4. _____ the food chain provides _____ to all living things.

a. extinction

b. hunt

c. energy

d. pesticides

1. _____ chemicals used to kill insects.
d

2. _____ when a link is broken, the animals above it are in danger of what?
a

3. _____ when we _____ animals nearly to extinction, we are putting the food chain in danger.
b

4. _____ the food chain provides _____ to all living things.
c

a. extinction

b. hunt

c. energy

d. pesticides

P	Q	W	E	R	T	Y	U	I	O	P	O
F	E	G	H	J	K	L	F	O	O	D	R
D	S	S	A	G	A	Z	X	C	V	B	G
G	D	H	T	N	K	N	K	M	M	N	A
F	A	D	S	I	A	U	I	O	K	J	N
Q	N	W	E	T	C	R	T	M	Y	U	I
L	G	K	J	N	H	I	J	H	A	G	S
Z	E	X	C	U	A	V	D	B	N	L	M
M	R	N	B	H	I	V	C	E	X	Z	S
A	S	D	F	G	N	H	J	K	S	K	L
O	I	E	N	E	R	G	Y	I	U	Y	T

Word Bank

pesticides	**animals**	**organisms**
hunting	**danger**	**chain**
food	**energy**	

Human Effects on the Food Chain

Find the words in the word search.

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